

Abstract Linear Algebra

Prof. Lanier

jlancier8@uchicago.edu

Pick up a handout. Try those warm-up problems.

Next up:

- Today: Chapter 2.1-3 of *LADW*
- HW₁ due tonight, 11:59pm. LaTeX template available on Canvas.
- Be sure to submit a homework reflection and response form.
- Computer Homework 1 on WeBWorK due on Saturday at 11:59pm.
- Quizzes 1 & 2 today at end of class.
- HW₂ & Homework 2 distributed today, due next Friday and Saturday.
- Problem session today, 3-5pm in Ryerson 177.
- Homework debrief on Sunday in this room, 3pm-4pm.

2.I

Warm-up 1

a) Solve the following system of equations over $\mathbb{R}$:

$$3x + 2y = 14$$

$$7x + 4y = 26$$

b) Write the vector $\begin{pmatrix} 14 \\ 26 \end{pmatrix}$ as a linear combination of $\begin{pmatrix} 3 \\ 7 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 4 \end{pmatrix}$ over $\mathbb{R}$.

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vector equation

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matrix equation

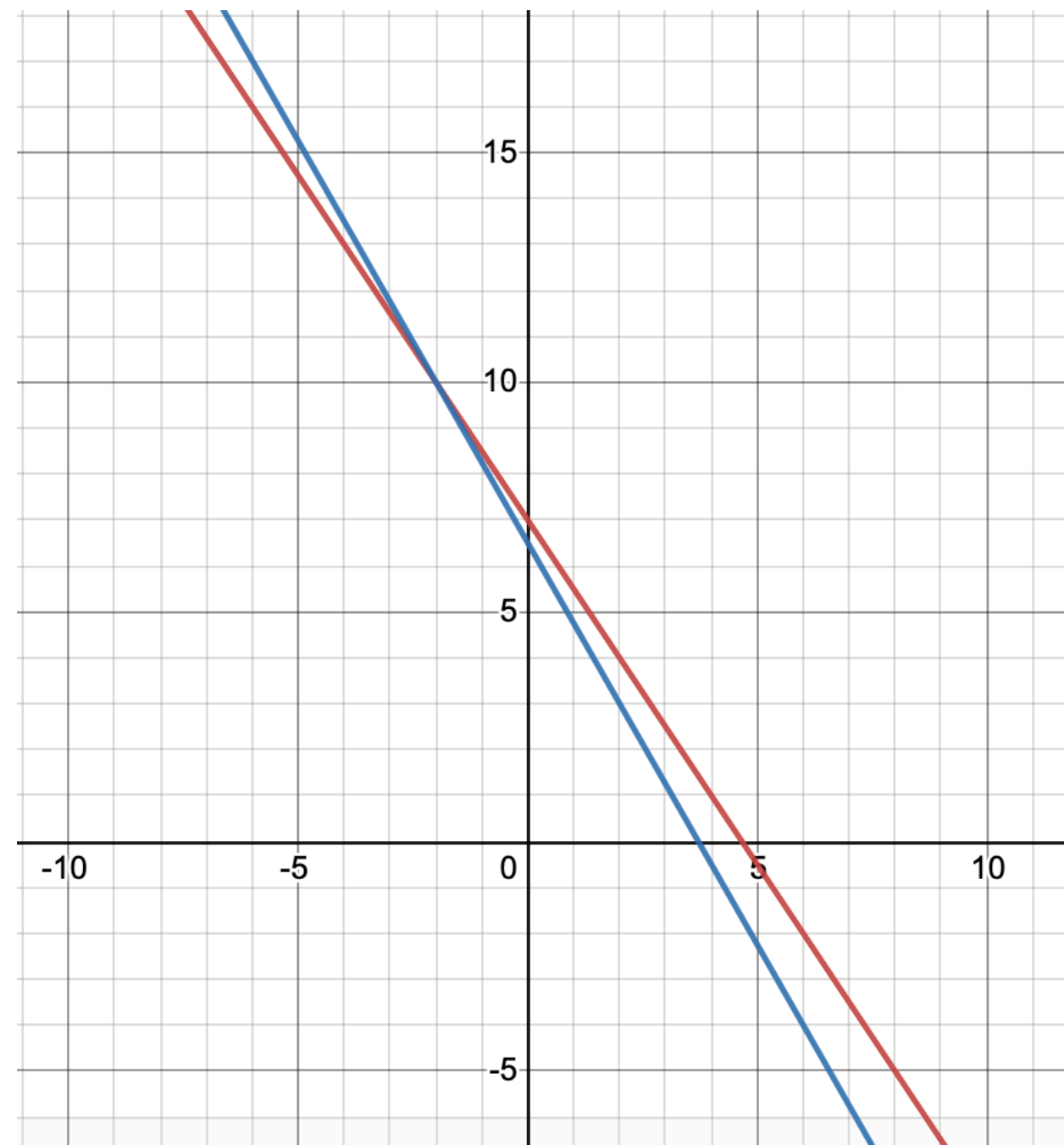
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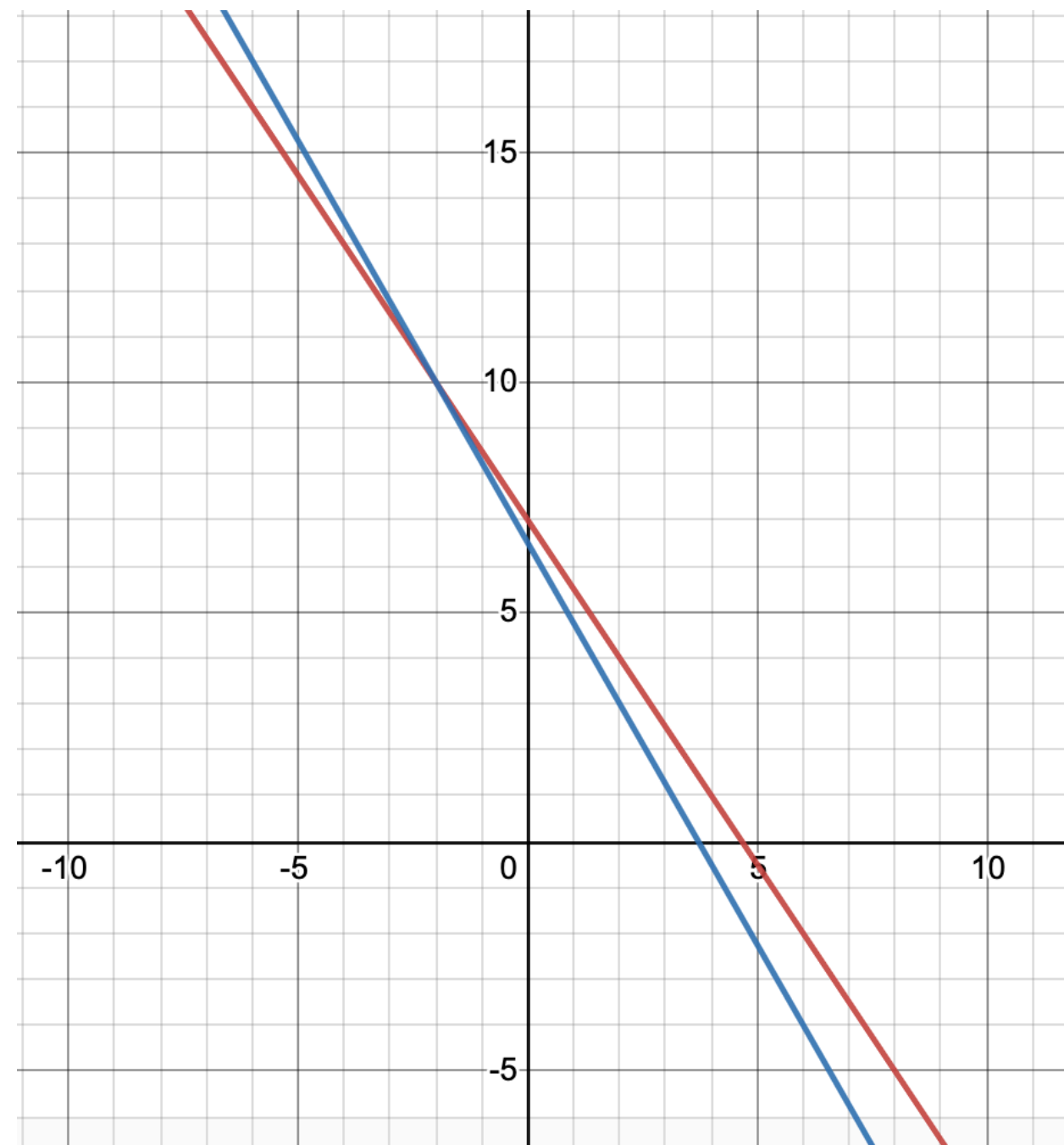
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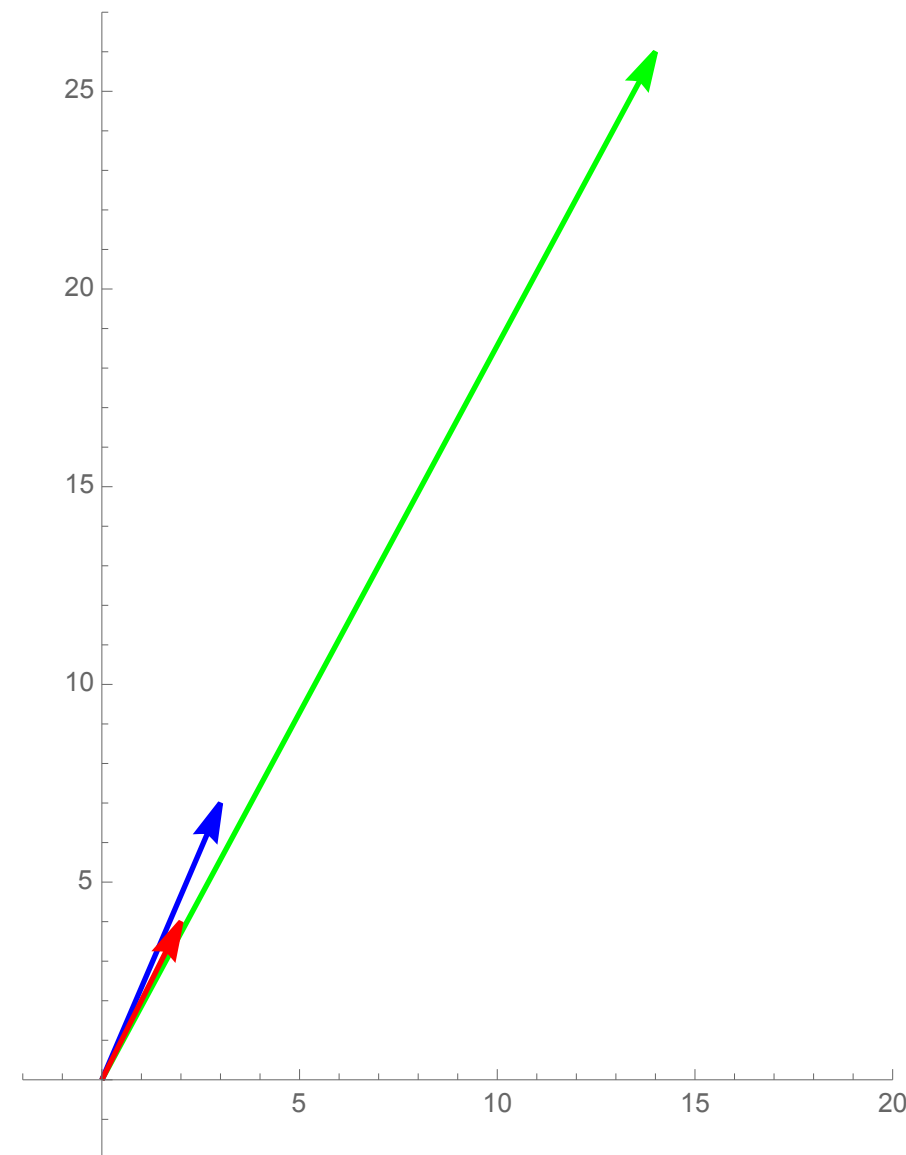
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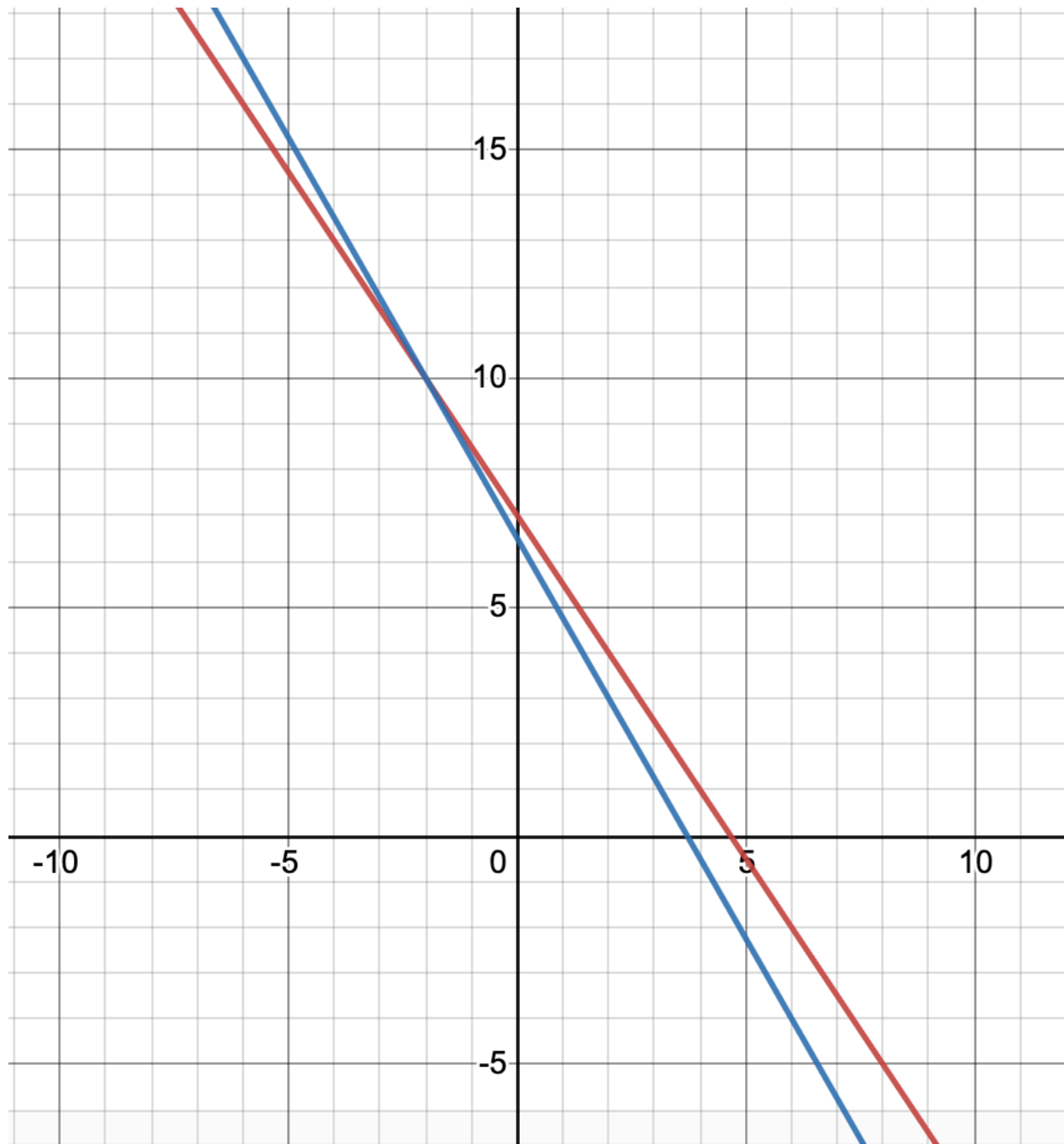
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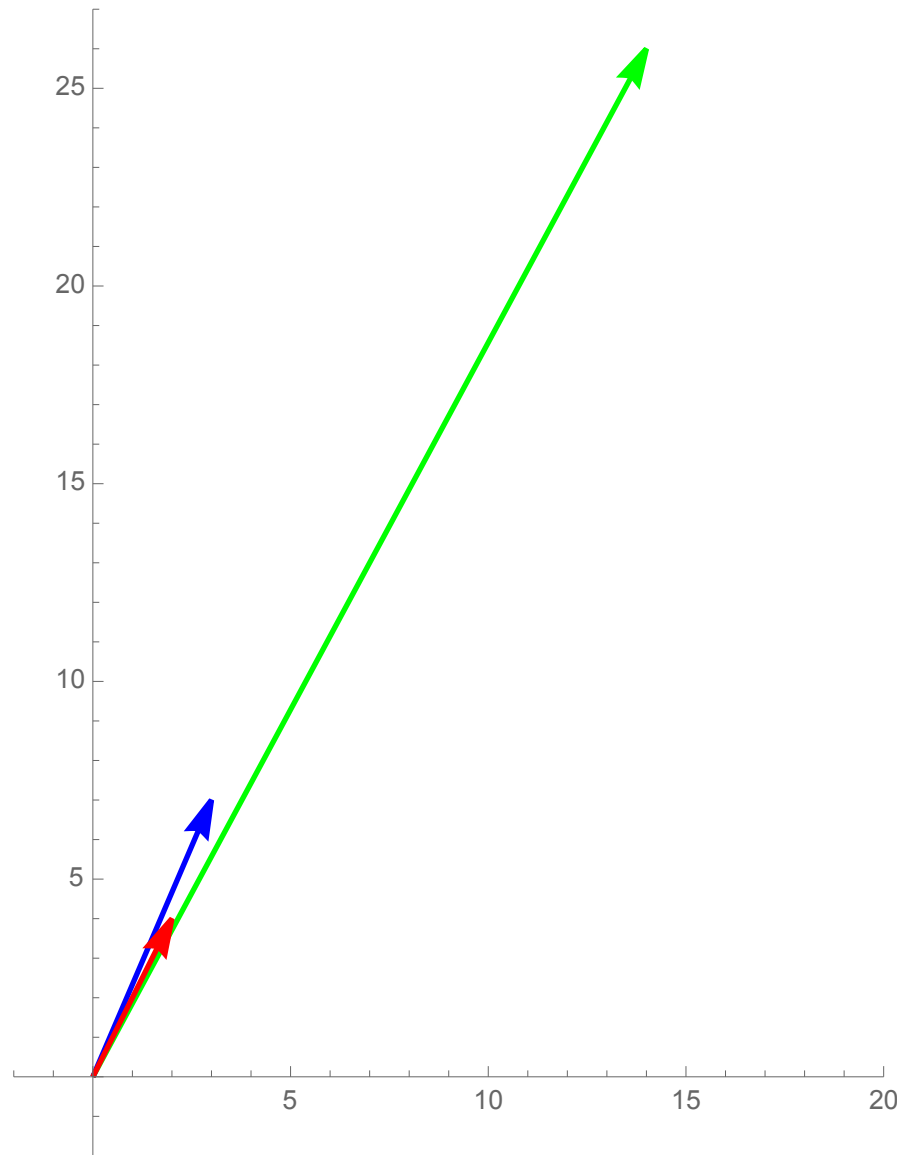
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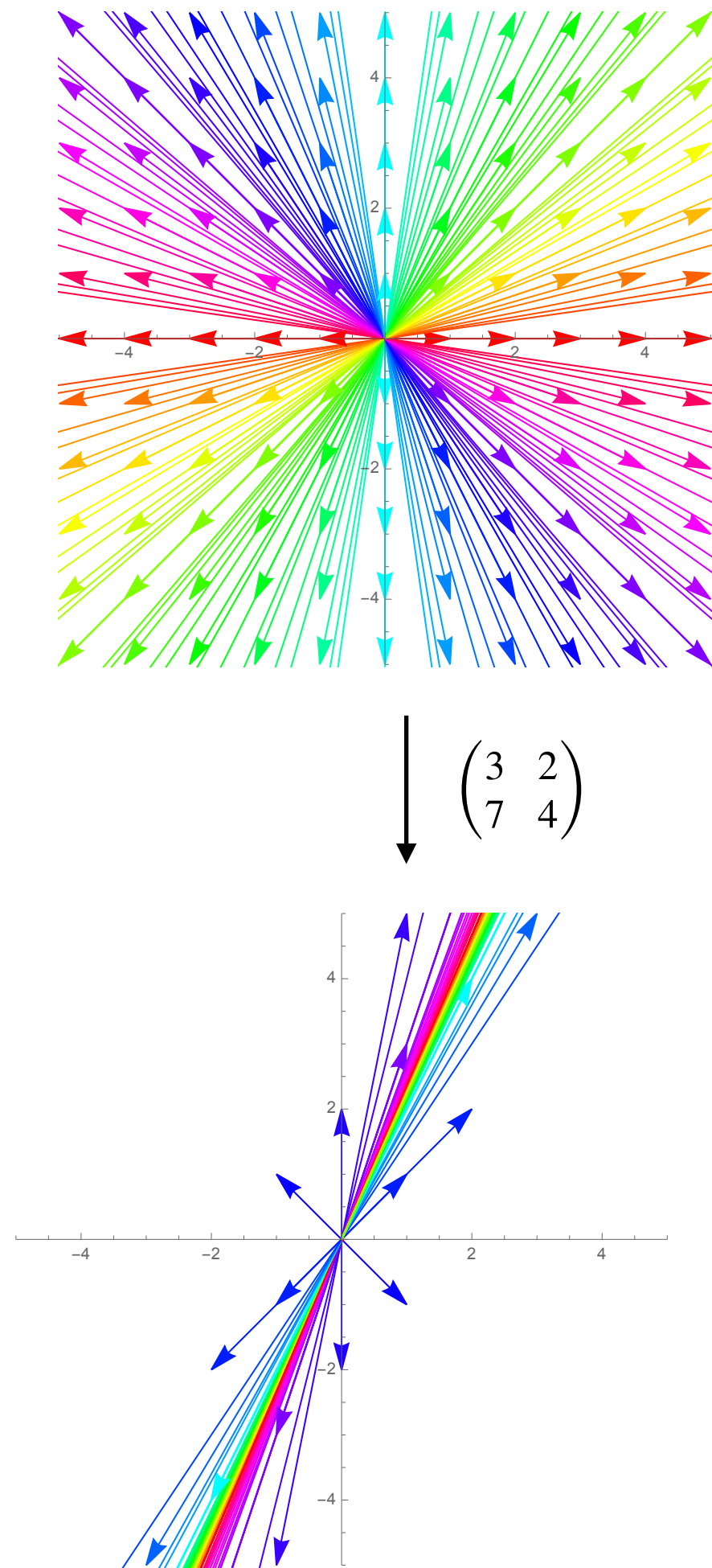
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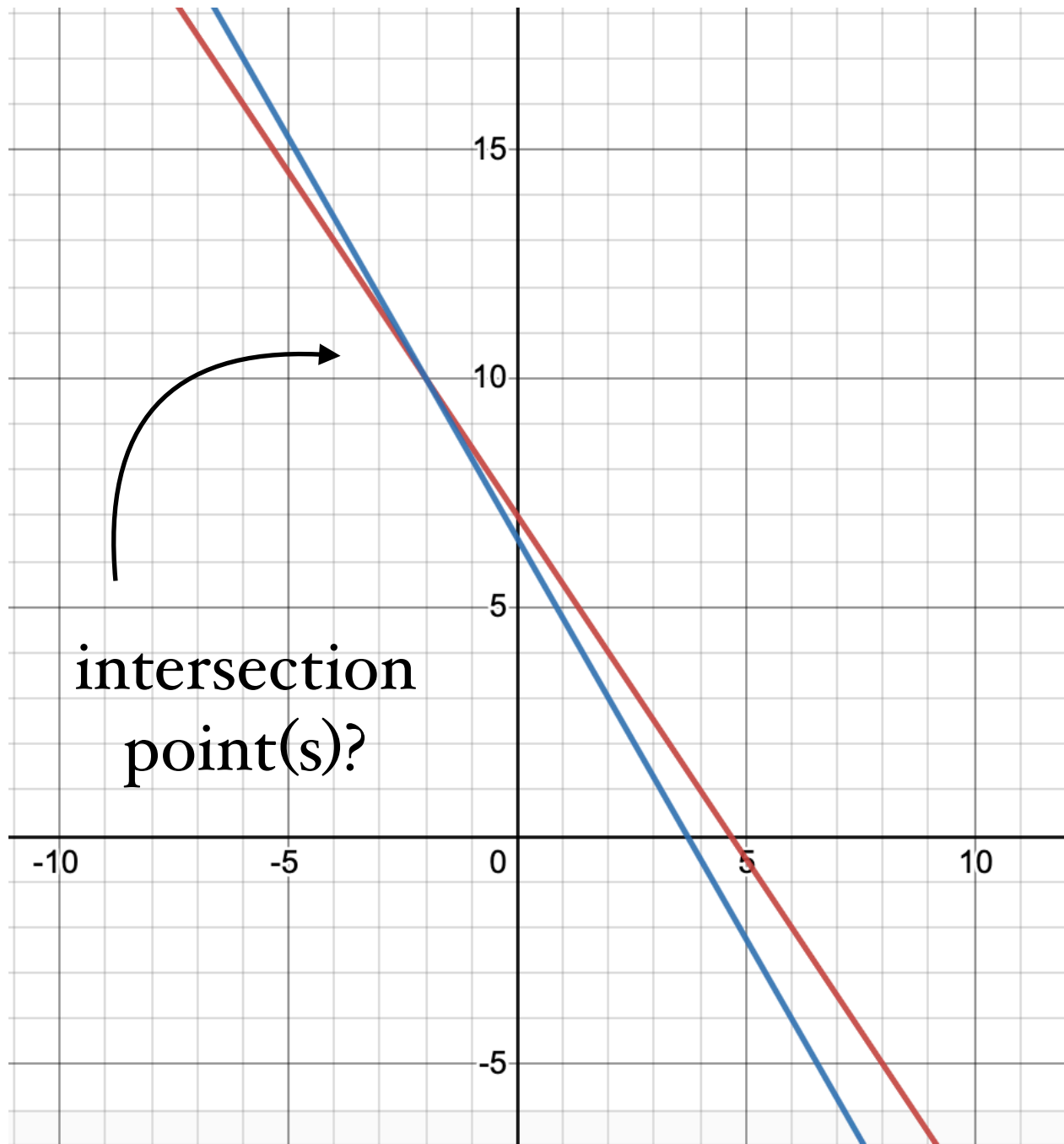
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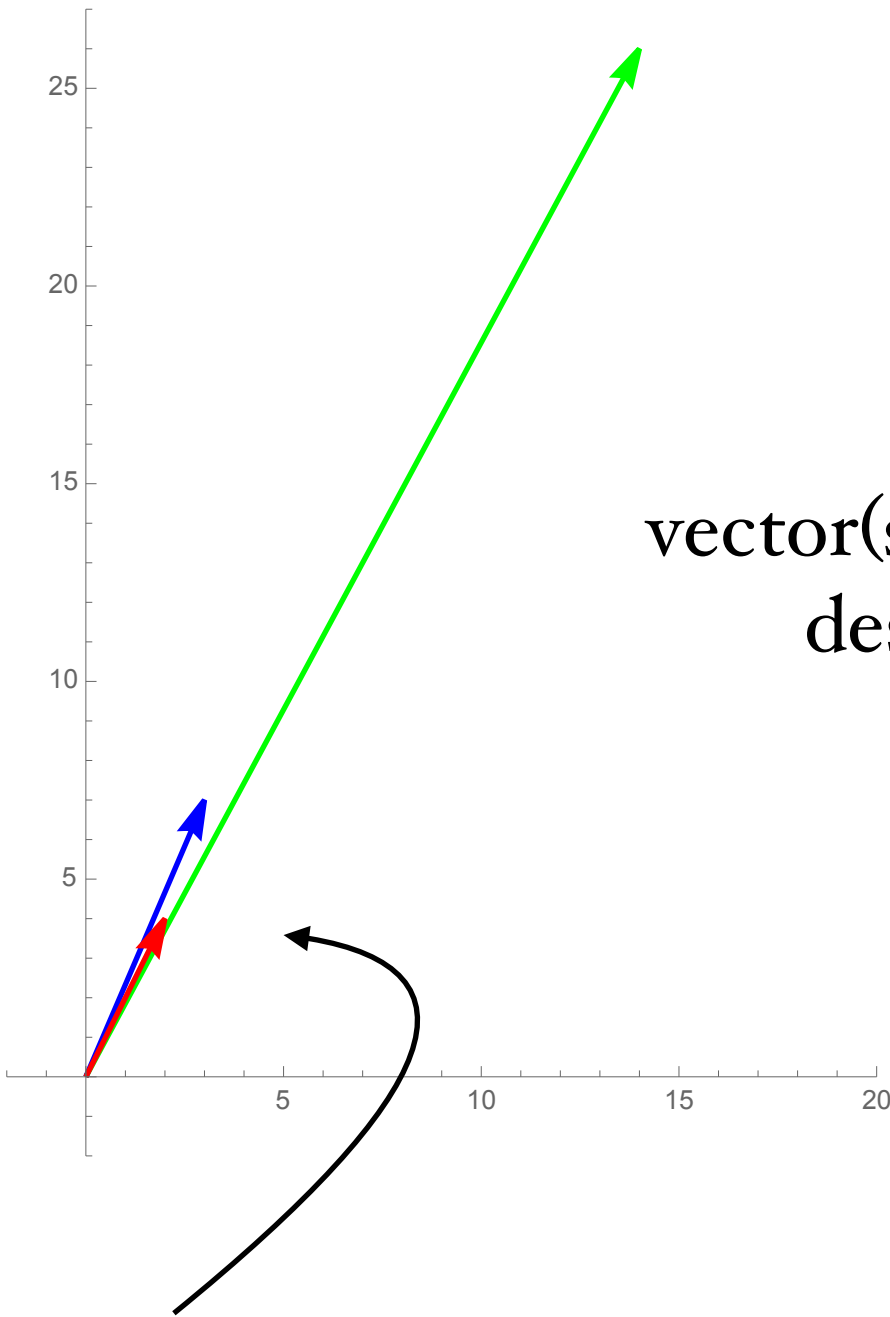
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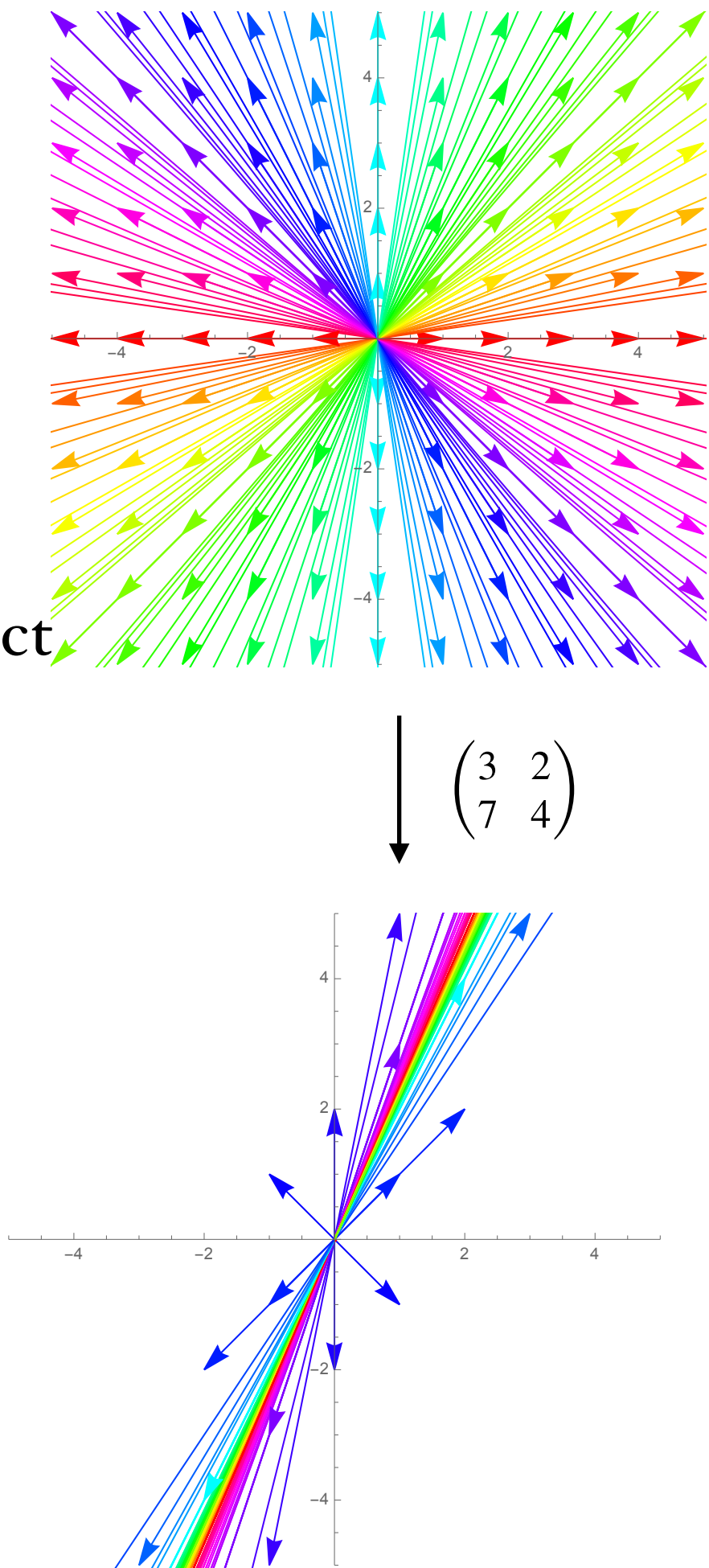
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tuple(s) of scalars for linear
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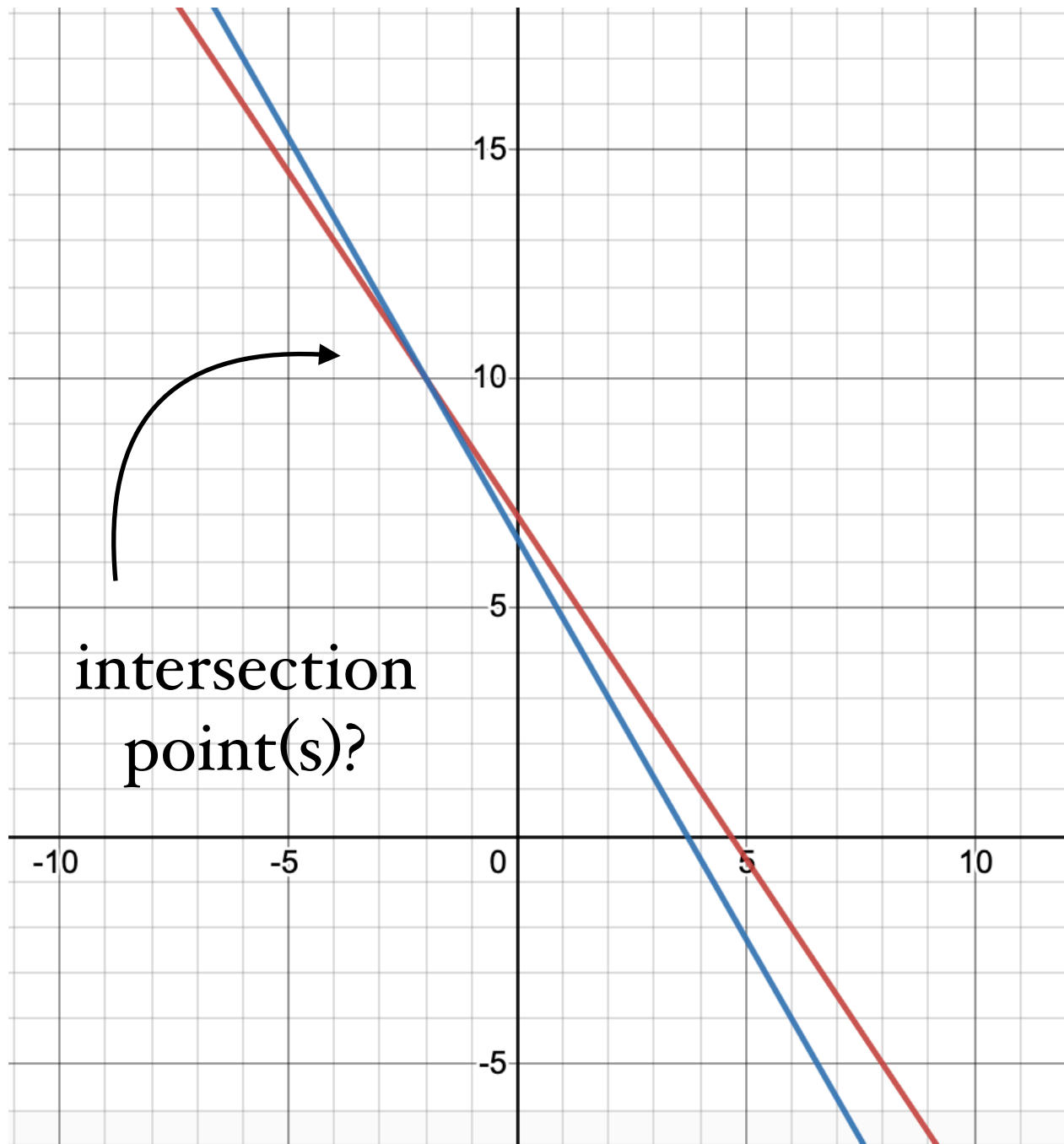


vector(s) with correct
destination?

2.I

system of
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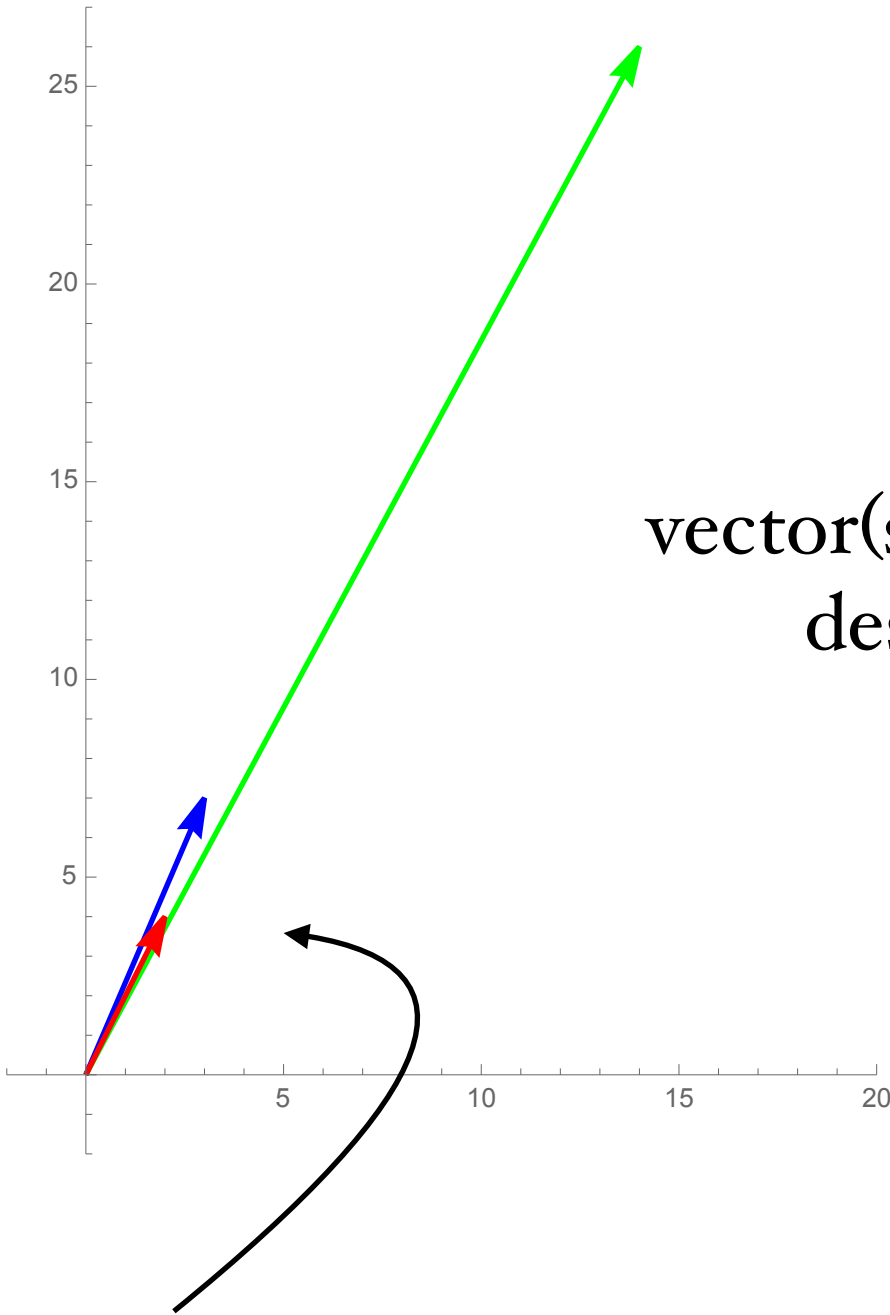
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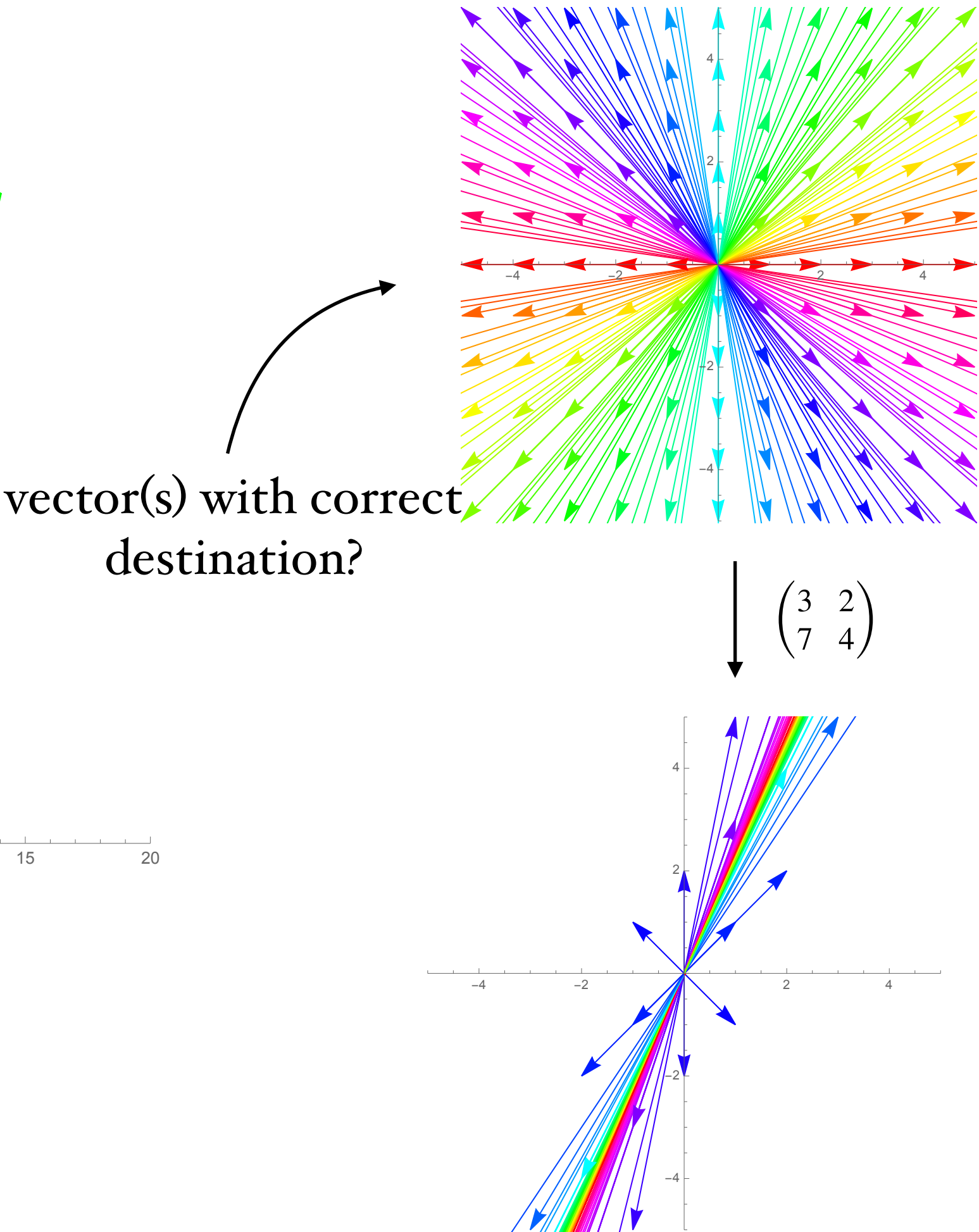
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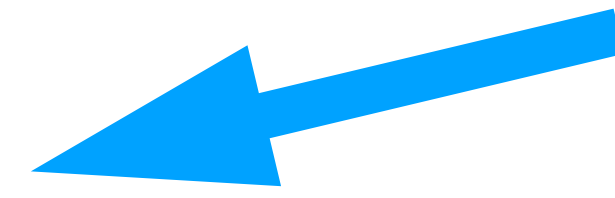
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I.7

The solutions to a (homogeneous) system
of linear equations form a vector space!

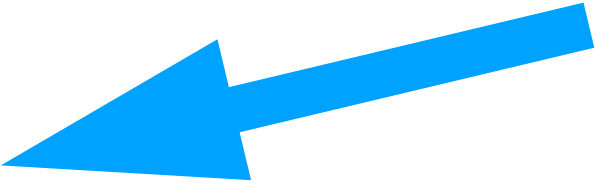


b/c need to have
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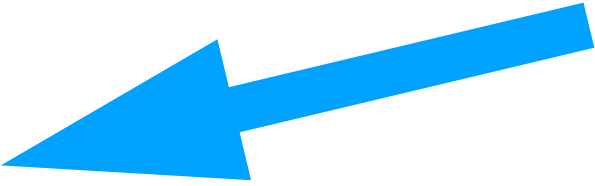


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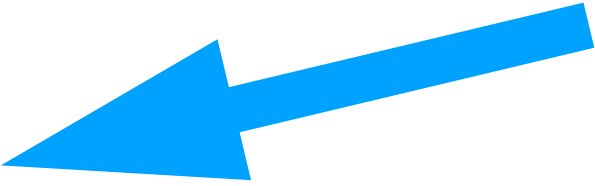
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For a homogeneous linear system with m variables and n equations,
the solution set forms a **subspace** of $\mathbb{F}^m$.

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b/c need to have the zero vector



The solutions to a homogeneous system of linear equations form a **subspace** of possible tuples.

For a homogeneous linear system with m variables and n equations, the solution set forms a **subspace** of $\mathbb{F}^m$.

This is called the nullspace.

Note: a homogeneous linear system always has at least one solution, the trivial solution.

row reduction

(elimination method, Gaussian elimination, Gauss-Jordan elimination)

$$3x + 2y = 14$$

$$7x + 4y = 26$$



$$\left(\begin{array}{cc|c} 3 & 2 & 14 \\ 7 & 4 & 26 \end{array} \right)$$

Use a sequence of (reversible) “moves” to put the system into a form where it is easy to read off the solutions.

Moves (row operations):

1. swap two rows
2. multiply a row by a non-zero scalar
3. add a multiple of a row to a row

Forms:

1. (row) echelon form
2. reduced (row) echelon form

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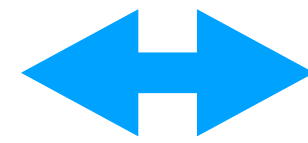
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Notice: these moves preserve the span of the row space!
But change the span of the column space!

Row equivalent matrices have the same row span,
but not necessarily the same column span.

$$\begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}$$

$R2 - 2R1$

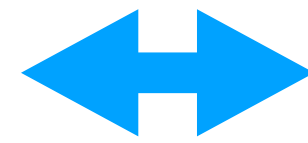


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row span: $\text{span}(\{(1 \ 1), (2 \ 2)\})$

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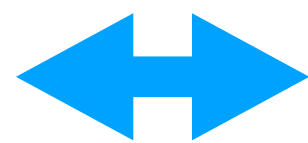
column span: $\text{span}(\left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\})$

$\text{span}(\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\})$

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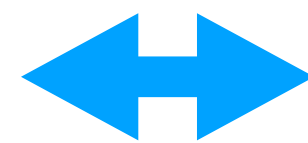
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$=$

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column span:

$$\text{span}\left(\begin{pmatrix} 1 \\ 2 \end{pmatrix}\right)$$

$\neq$

$$\text{span}\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right)$$

matrix:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 8 \end{pmatrix}$$

row echelon form?

reduced row echelon form?

pivot: first non-zero entry in a row

matrix:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 8 \end{pmatrix}$$

row echelon form?

each pivot to the right of those above?
zero rows at bottom?

reduced row echelon form?

additionally:
all pivots equal 1?
above each pivot, all zeroes in its column?

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$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \end{pmatrix} \xrightarrow{R1 - 2R2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \end{pmatrix}$$

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During class 1

For each of the following matrices, decide: Is it in row echelon form? is it in reduced row echelon form? When the answer is yes, put a check. When the answer is no, apply row reduction to achieve these forms.

matrix:

$$\begin{pmatrix} 0 & 1 & 2 & 0 & 3 & 4 \\ 0 & 0 & 0 & 1 & 0 & 3 \end{pmatrix}$$

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$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix}$$

Why do we care about these forms?

REF

RREF

$$\left(\begin{array}{cc|c} 3 & 2 & 14 \\ 7 & 4 & 26 \end{array} \right)$$

$$\left(\begin{array}{cc|c} 1 & 1 & 28 \\ 0 & 2 & 20 \end{array} \right)$$

$$\left(\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & 10 \end{array} \right)$$

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not yet solved

ready to
back substitute

fully solved

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